TRAN THI THANH HOA

Human Factors / Human–Machine Interaction Engineer
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SUMMARY

PhD Researcher in Human–Machine Interaction with experience studying how user interact with complex technical systems. Skilled in usability studies, and analysis of user behavior in immersive and simulated environments. Experienced in 3D modelling environments, virtual reality simulation, and prototyping to study operator interaction with systems and interfaces. Strong background in multidisciplinary collaboration with engineering teams to translate user analysis into system improvements.
Published researcher in international conferences (IEEE VR, ISMAR, CAD). Fluent in English and French.

SKILLS

- **Human Factors & Ergonomics**: Human–machine interaction analysis, user behavior analysis in simulated environments, usability testing and experimental design, user interviews and surveys, eye-tracking studies (Tobii)
- **3D Simulation & Prototyping:** VR/AR simulation environments, 3D modelling and interaction design, SolidWorks, Creo, Blender, Unity
- **Programming & Data Analysis**: MATLAB, Python, R, C++, C#, GitHub
- **Collaboration & Tools**: Miro, Redmine, Visual Studio, Git
- **Languages**: English (fluent), French (fluent), Vietnamese (native)

PROFESSIONAL EXPERIENCE

PhD Researcher – Human-Machine Interaction

IMT Atlantique, Brest, France & Adelaide University, Australia | *09/2022 – 10/2025*

Research focused on **human interaction with autonomous vehicle** systems in simulated environments.

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- Applied **qualitative and quantitative human factors methods** including interviews, surveys, and eye-tracking analysis.
- Developed a **VR-based driving simulator to prototype and evaluate new visual interaction concepts**.
- Analyzed **user behavior and interaction patterns** to identify usability issues and improve system design.
- Published and presented results at **IEEE VR, ISMAR, ICAT.**

Tools: Unity, C#, Git, Python, R, Blender, Tobii Eye Tracker, Quest Pro.

R&D Engineer – C++ / Virtual Reality

TechViz, Paris, France | *02/2022 – 08/2022*

Worked with engineering teams developing application tools for test automation

- Collaborated with engineering teams to developed and improved an automated testing platform.
- Collected and analyzed user feedback to enhance usability and workflow efficiency.

Tools: C++, C#, JavaScript, Git, Redmine, Bash, HTC Vive, Valve, CAVE, Creo.

Research Engineer – Graduation Project
CNRS, Grenoble, France | *02/2021 – 08/2021*

Developed a **virtual reality system for 3D modelling and interaction with geometric objects**.

- Designed an **immersive VR modeling system** supporting direct 3D modeling operations (resize, translate, rotate, Boolean operations).
- Oversaw the entire development cycle: concept, prototype, user study, and publication
- Presented outcomes at the **CAD 2021 conference**.

Tools: Unity, C#, Git, Python, SteamVR, FreeCAD, HTC Vive.

Team Leader – AVATAR Project (Erasmus+)
Grenoble, France | *02/2021 – 06/2021*

- Led a team of 5 students from 4 European universities on a **digital twin VR simulation project**.
- Managed project timeline, tasks, code reviews, and cross-cultural collaboration
- Contributing to an open-source 3D modeling program in VR, code reviews, and commits.

Tools: Unity, Git, Creo, ROS, MATLAB.

EDUCATION

PhD in Computer Science (Human-Machine Interaction)
IMT Atlantique, France & Adelaide University, Australia | *2022 – 2025*
France–Australia Cotutelle Program.

Engineer's Degree – Product Engineering
Grenoble INP, France | *2018 – 2021*
Awarded **Eiffel Excellence Scholarship** from the French Ministry of Europe and Foreign Affairs.

Engineer's Degree – Automated Production Engineering
Danang University of Technology, Vietnam | *2015 – 2018*
Top 1 of 80 students.

CERTIFICATIONS

- Advanced Virtual and Augmented Reality Toolkit for Learning – Erasmus Mundus
- 3D Interaction Design in Virtual Reality – Coursera ID: 58YVZLJ89MV2
- Building 3D Interactive and Social VR – Coursera ID: LMPVNEAREDMS

PUBLICATIONS

- *Direct Solid Modelling in Virtual Reality for Systems Architecting*.
- *Benchmarking of 3D Modelling in Virtual Reality, CAD 2021 Conference*.
- *Impact of Adding, Removing and Modifying Driving and Non-Driving Related Information on Trust in Autonomous Vehicles.* IEEE VR.